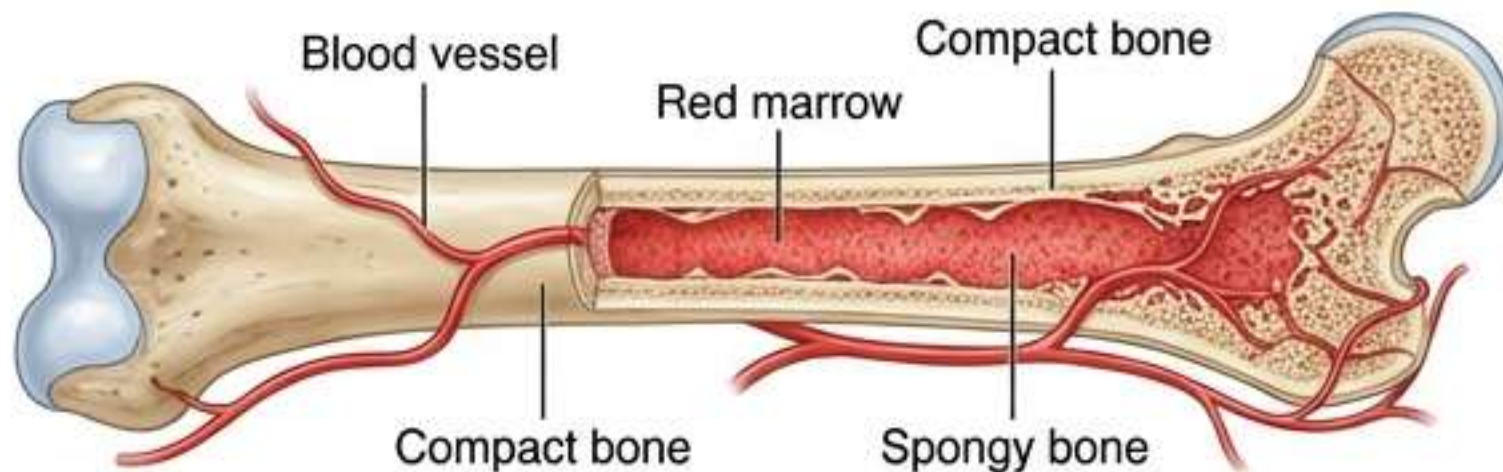


QUESTION 01 | SOURCE: EMD 1-2023 (Q06)

Regarding the skeletal system: Bone is a:

- A. Highly vascularized connective tissue
- B. Elastic connective tissue
- C. Organ responsible for manufacturing blood cells
- D. Rigid non-innervated organ
- E. Phosphocalcic reservoir

Correct Answer: A, C, E



Bone is a specialized, rigid **connective tissue** that is **highly vascularized** (unlike cartilage, which has no vessels). It acts as a dynamic **mineral reserve** (calcium and phosphorus) and contains red marrow responsible for **hematopoiesis** (producing blood cells). It is definitely innervated.

Vascularized, **R**eservoir, **P**roducer.

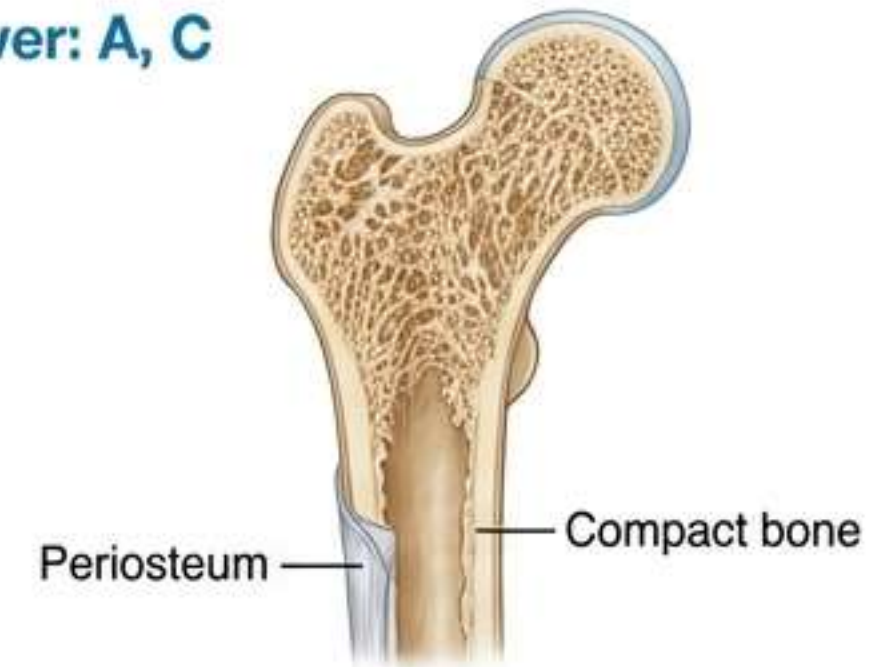
Remember **VRP** (Very Rigid & **P**roductive).

QUESTION 02 | SOURCE: EMD 1-2023 (Q07)

Regarding the skeletal system: Bone is formed by:

- A. A fibro-elastic membrane called periosteum
- B. Compact tissue forming the entirety of the bone
- C. Spongy tissue occupying the epiphyses of long bones
- D. Periosteum covering the articular surfaces
- E. Soft and impermeable connective tissue

Correct Answer: A, C



Bone structure is composite. The **periosteum** is the outer fibrous membrane, but it *never* covers articular surfaces (cartilage does that). **Spongy bone** is found internally, specifically filling the **epiphyses** of long bones, while compact bone forms the outer shell.

Spongy centers, **Compact** coats.

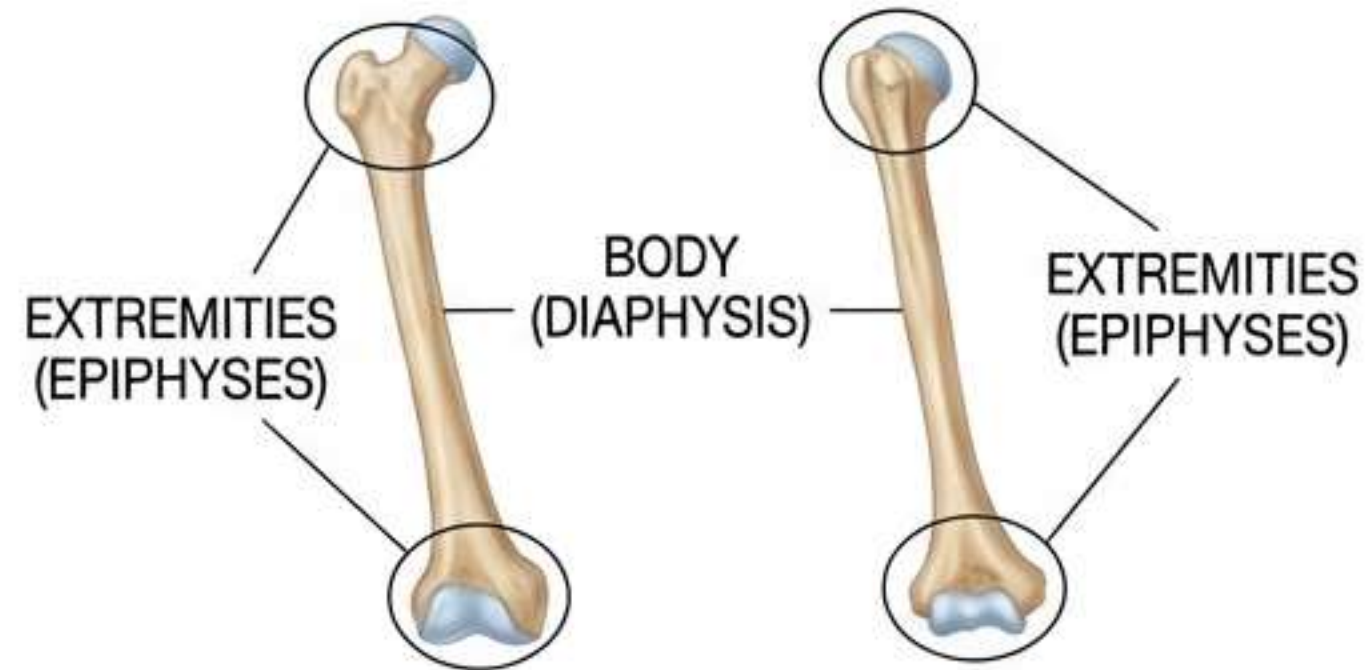
Periosteum = **Perimeter** (except at the joints).

QUESTION 03 | SOURCE: EMD 1-2023 (Q08)

Regarding the skeletal system: The long bone:

- A. Constitutes the entire skeleton of the limbs
- B. Presents a body and two extremities
- C. Consists of a smooth surface throughout
- D. Is always articular at its extremities
- E. Is covered with hyaline cartilage over its entire surface

Correct Answer: B, D



A long bone is defined by its morphology: a shaft (**diaphysis/body**) and two ends (**epiphyses/extremities**). The **epiphyses** have **articular surfaces** to form joints. The bone surface is *not* smooth everywhere; it has protrusions (tuberosities/crests) for muscle attachment.

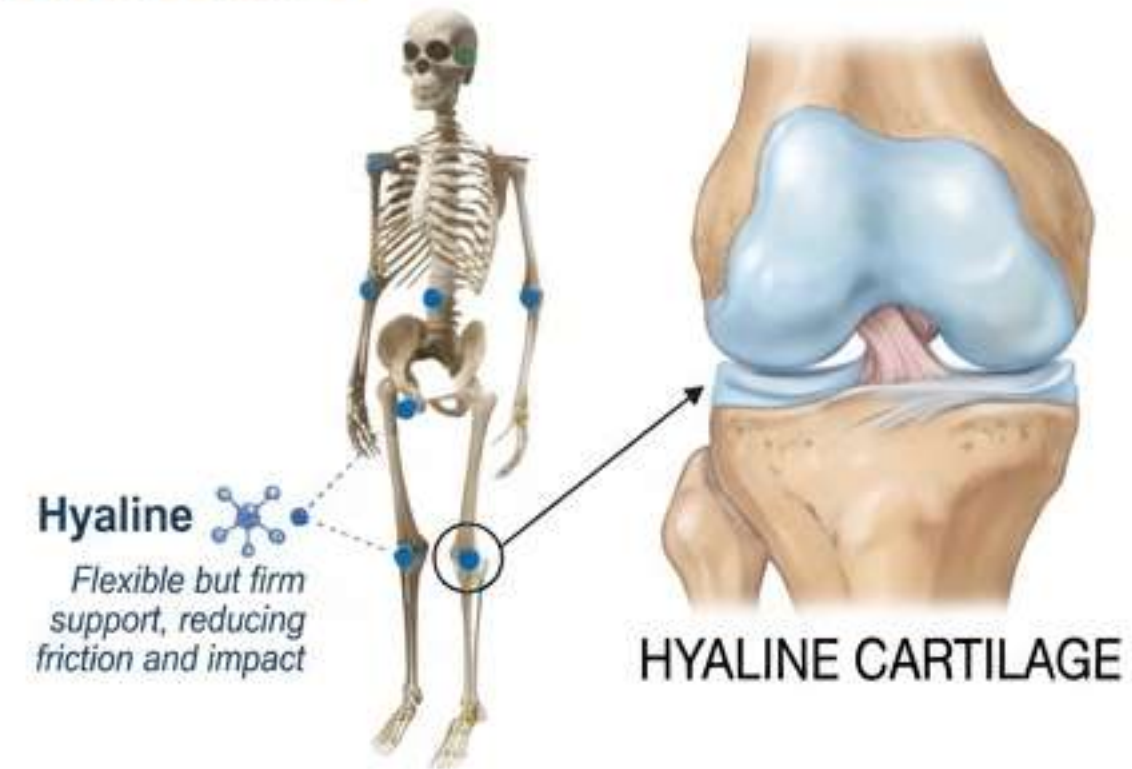
Long bone = **D**iaphysis + 2 **E**piphyses. Think **D-E-E**.

QUESTION 04 | SOURCE: EMD 1-2023 (Q09)

Concerning osteology:

- A. A flat bone has a diaphysis and two epiphyses
- B. The articular surface is covered by articular or hyaline cartilage
- C. Red bone marrow is found in the diaphysis of long bones (in adults)
- D. The surface of bones is always regular
- E. There is only one type of bone

Correct Answer: B



Hyaline cartilage is the most common type and coats articular surfaces to reduce friction. Flat bones do not have diaphyses/epiphyses (those are terms for long bones). In adults, the diaphysis contains **yellow** (fatty) marrow, not red.

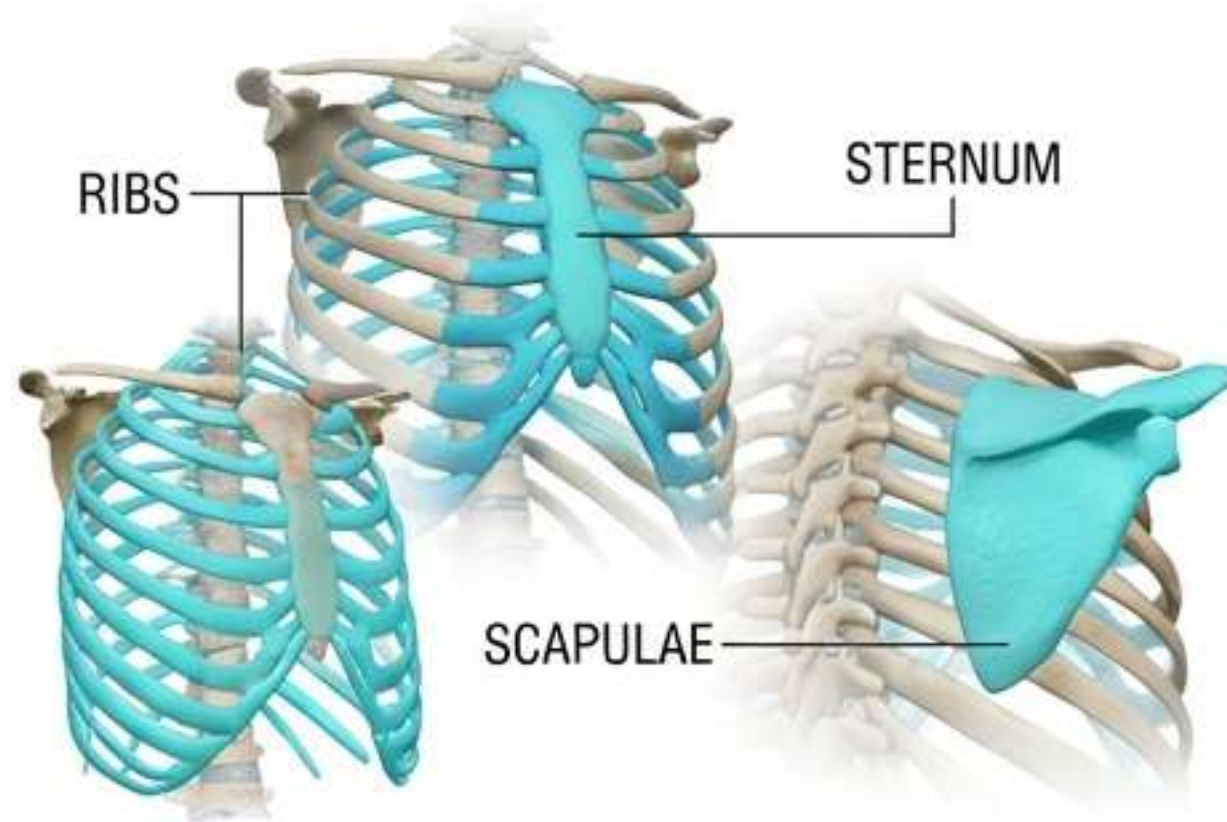
Hyaline = **High-Shine**. Smooth and glassy for joints.

QUESTION 05 | SOURCE: EMD 1 2022 (Q04)

Concerning the skeletal system:

- A. It is an important reserve of minerals
- B. Manufactures blood cells
- C. Stores heavy materials
- D. Protects organs
- E. Provides energy to the body

Correct Answer: A, B, C, D



The skeletal system is a multitasker: **Protection** (ribcage/skull), **Movement**, **Hematopoiesis** (blood cells), and **Mineral Storage** (Calcium/Phosphorus). Heavy metals (like lead) can accumulate in the bone matrix, making C technically correct in a pathological/storage context.

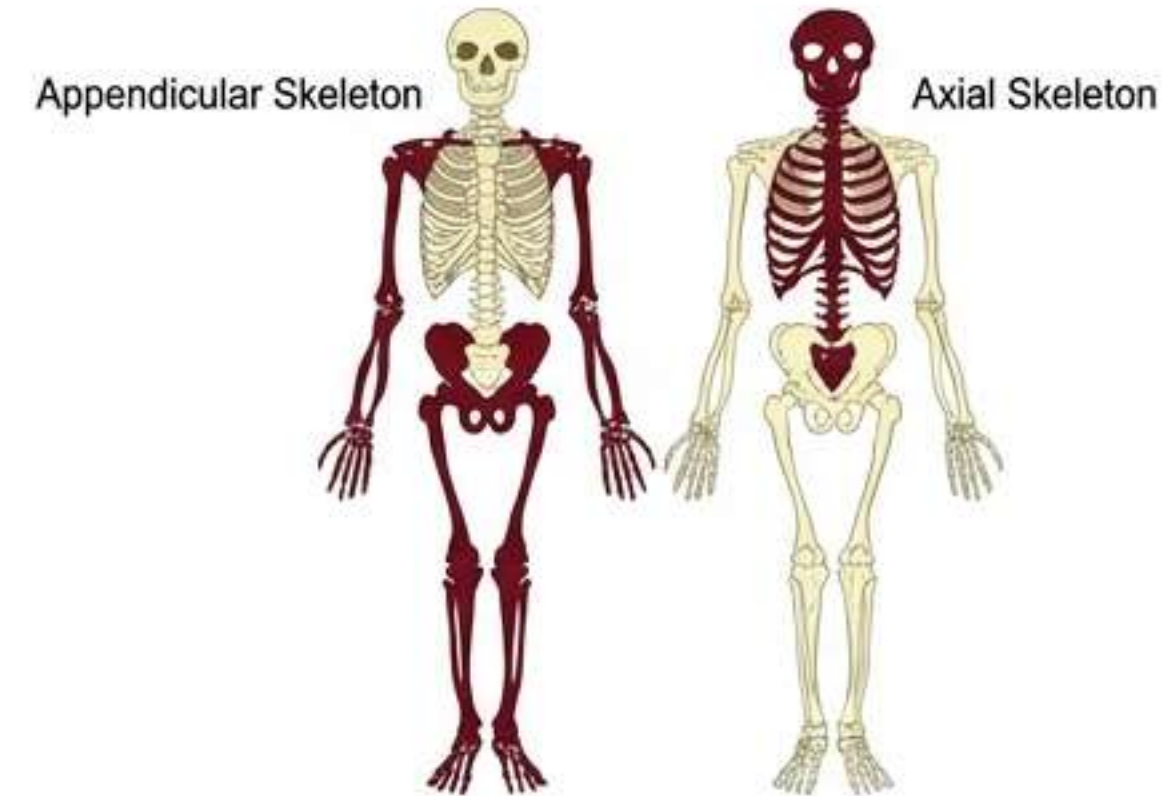
P.M.S. functions: **P**rotection, **M**ovement, **S**torage (and Synthesis).

QUESTION 06 | SOURCE: EMD 1 2022 (Q05)

Concerning the skeletal system:

- A. It represents 40% of body weight
- B. It plays a role in mobility
- C. Bone is formed by rigid connective tissue
- D. The axial skeleton is formed by the limbs
- E. The surface of the bone is always regular

Correct Answer: A, B, E (See Note)



CRITICAL EXAM NOTE: The lesson text states the skeleton is **20%** of body weight, but the exam key marks **A (40%)** as correct. Similarly, surfaces are irregular, but the key marks **E** (regular) as correct. **Strictly for the exam:** Select A, B, E.

Biologically: It is ~20% and irregular.

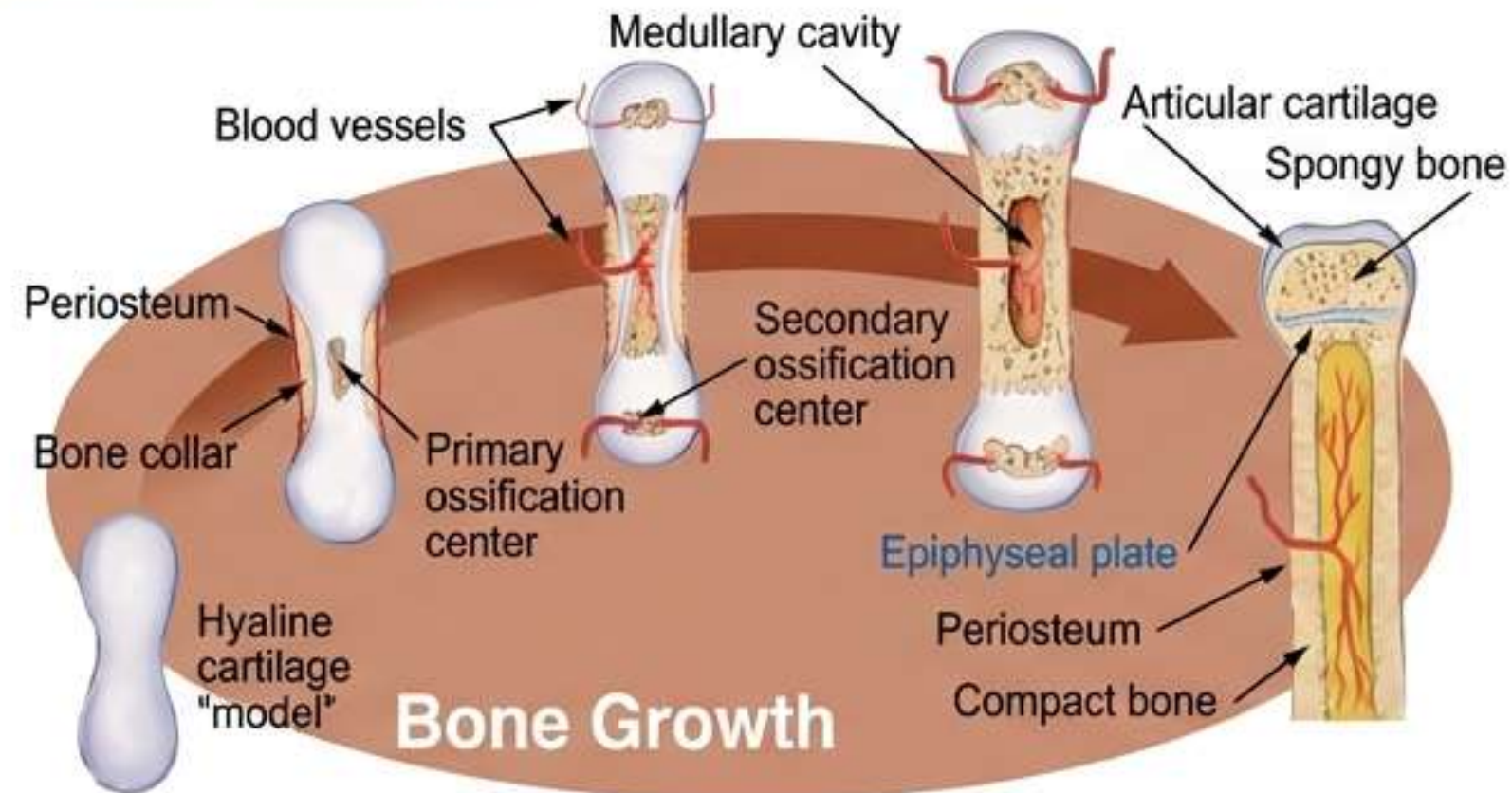
CAUTION: Memorize the **Key** (40%), but know the **Fact** (20%).
Exam logic ≠ Real Life. By dentiste web

QUESTION 07 | SOURCE: EMD 1 2022 (Q06)

Concerning the structure of bone:

- A. The diaphysis of the long bone is hollowed by the medullary canal
- B. The periosteum is a fibrous membrane covering the bone
- C. The osteoclast is the bone-forming cell
- D. The metaphysis is comprised between the epiphyses
- E. Growth of bone in length is ensured by the growth cartilage

Correct Answer: A, B, E



Professional Explanation:

Osteoblasts build bone; **Osteoclasts** destroy/resorb it (Statement C is false). The **medullary canal** runs through the center of the diaphysis. Length growth occurs at the **metaphysis** via the growth plate (cartilage).

Mnemonic:

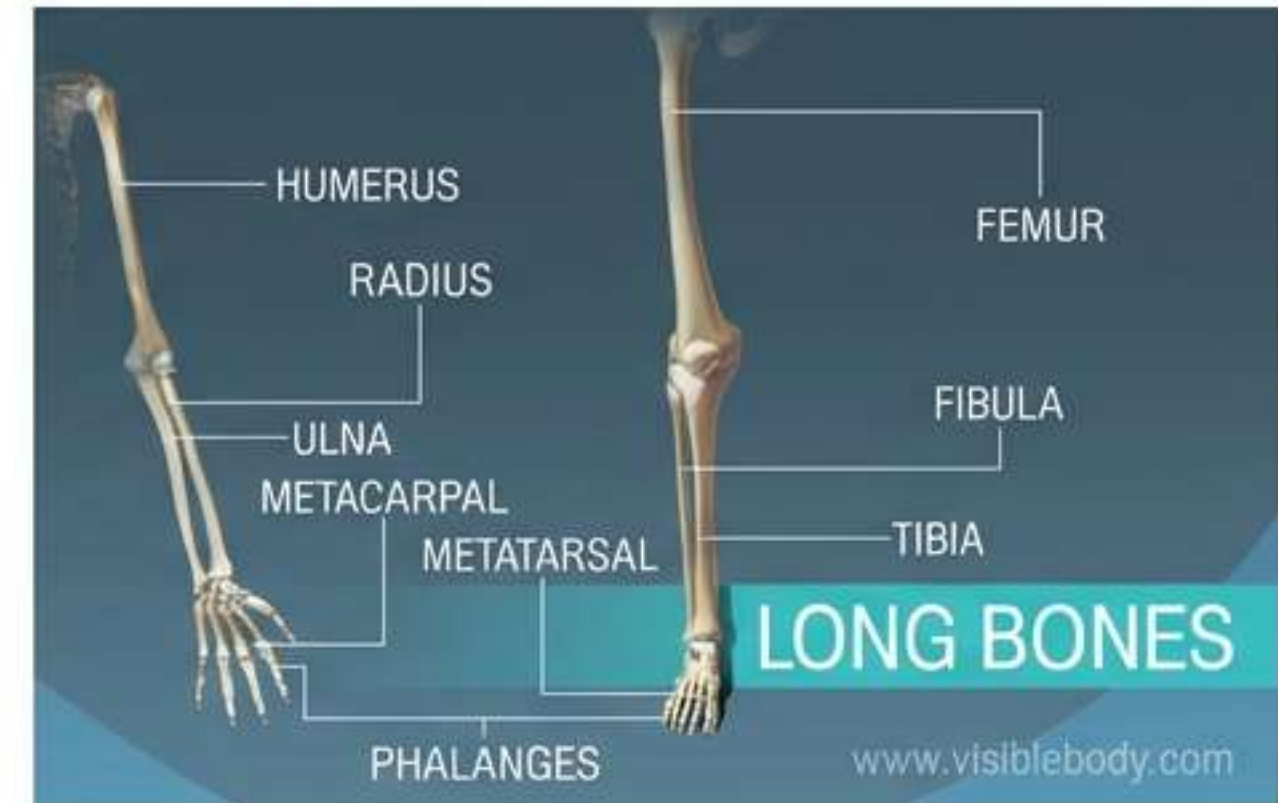
Blasts **B**uild. **C**lasts **C**rash (Consume).

QUESTION 08 | SOURCE: EMD 1 2022 (Q15)

A long bone is:

- A. Longer than it is wide
- B. Wider than it is long
- C. Thicker than it is wide
- D. Presents growth cartilage at the level of the diaphysis
- E. Presents articular cartilage at the level of the epiphyses

Correct Answer: A, E



Professional Explanation:

Classification is based on dimensions. **Long bones** (e.g., Humerus) have Length > Width/Thickness. Growth cartilage is located at the **metaphysis** (junction), not the center of the diaphysis. Articular cartilage always caps the **epiphyses**.

Mnemonic:

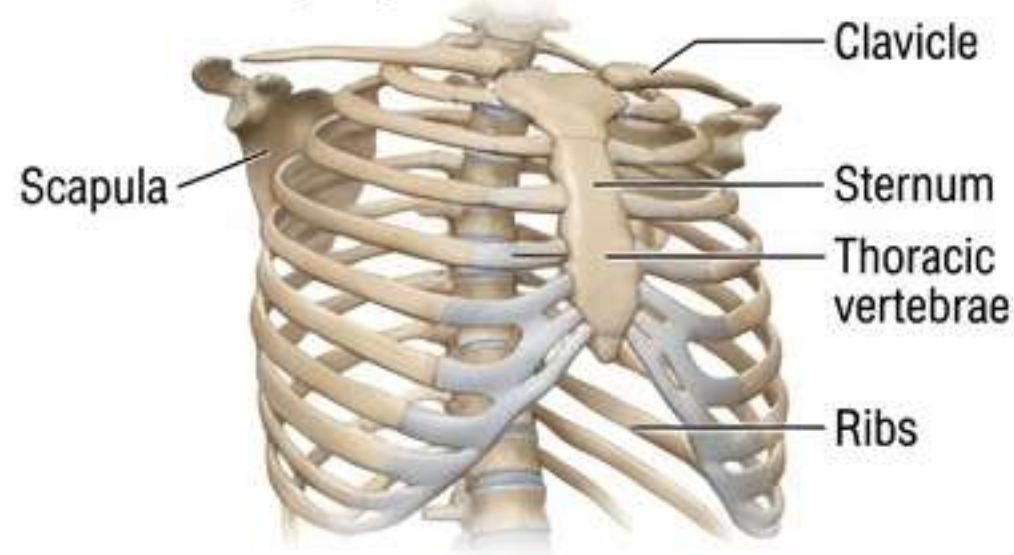
Long = **Length** leads. Growth happens at the **Junction** (Metaphysis).

QUESTION 09 | SOURCE: EMD 1 2021 (Q13)

The thoracic cage belongs to the axial skeleton. Which bones constitute it?

- A. The sternum
- B. The clavicle
- C. The femur
- D. The ribs
- E. The thoracic vertebral column

Correct Answer: A, D, E



The **Axial Skeleton** is the central axis. The thoracic cage specifically includes the **Sternum** (front), **Ribs** (sides), and **Thoracic Vertebrae** (back). The clavicle is part of the shoulder **girdle** (appendicular), not the cage itself.

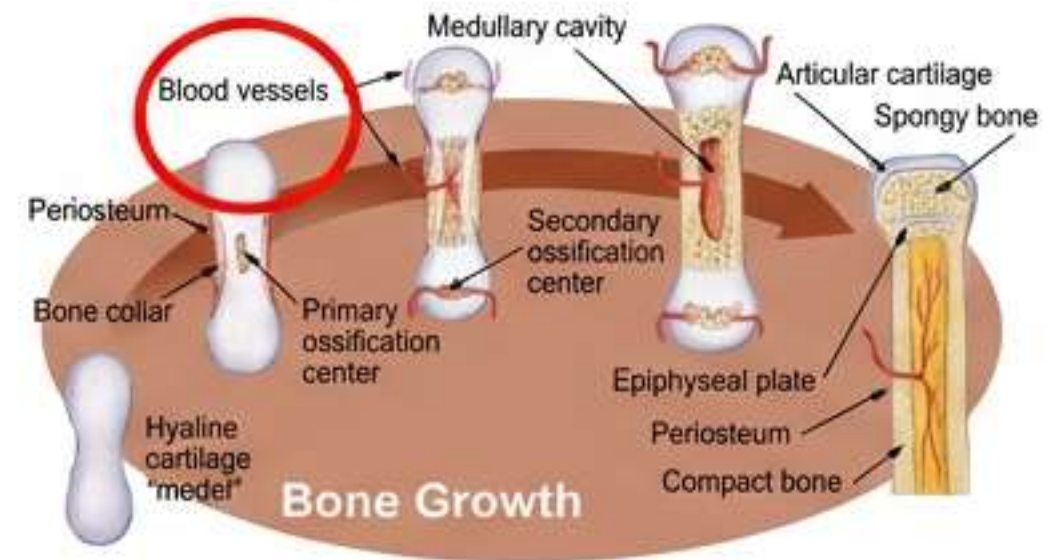
Cage Components: **S-R-V**. **S**ternum, **R**ibs, **V**ertebrae.

QUESTION 10 | SOURCE: EMD 1 2021 (Q15)

A long bone is:

- A. Longer than it is wide
- B. Wider than it is long
- C. Thicker than it is wide
- D. Vascularized by muscular arteries
- E. Contains growth cartilage at the level of the extremities

Correct Answer: A, E



Reiteration of long bone definition (Length > Width). Vascularization primarily comes from specific **nutrient arteries** entering the nutrient foramen. Growth cartilage is found at the **extremities** (metaphysis/epiphysis junction) in children.

Growth at the **Ends**. Nutrients enter the **Middle**.

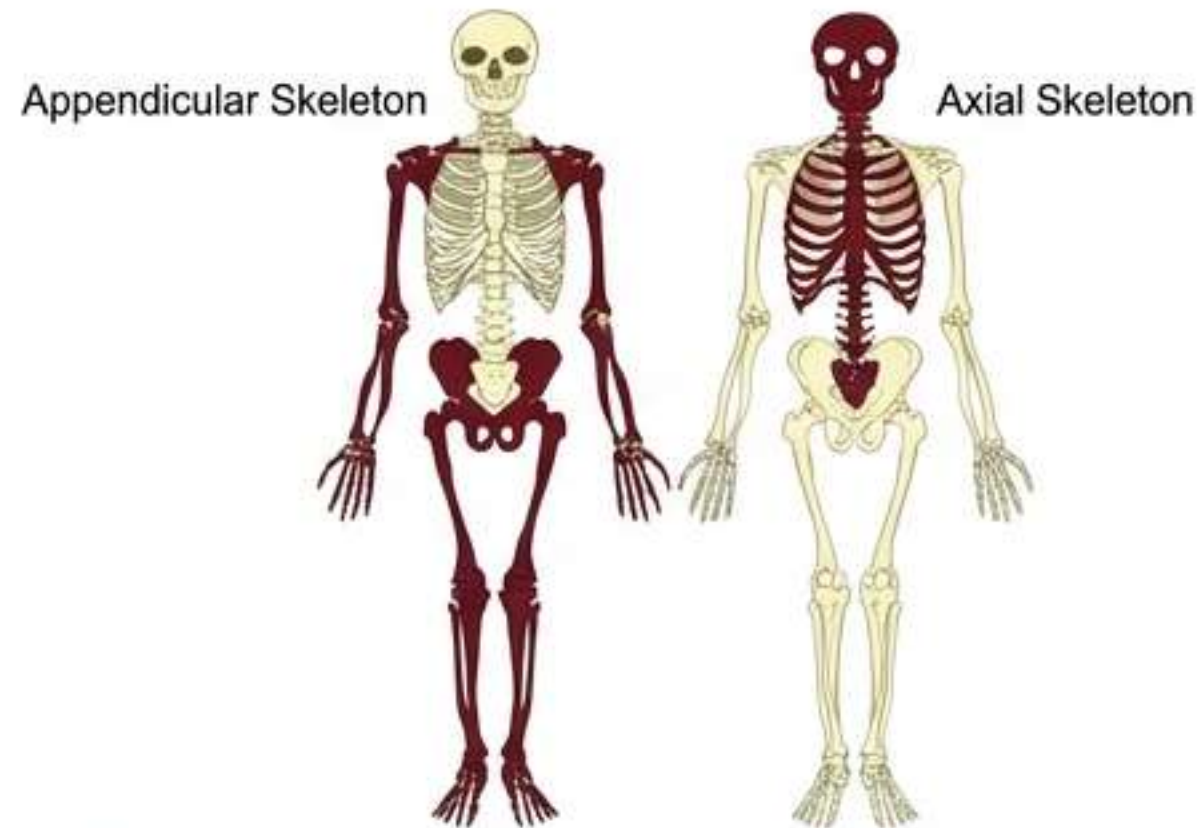
By dentiste web

QUESTION 11 | SOURCE: EMD 1 2020 (Q01)

About osteology:

- A. The appendicular skeleton is attached to the axial skeleton by girdles
- B. The femur is the longest bone in the human body
- C. The forearm skeleton is formed by a single bony piece, the radius
- D. The bones of the appendicular skeleton comprise articular surfaces
- E. The adult human body comprises 300 bones

Correct Answer: A, B, D



The appendicular skeleton (limbs) hangs off the axial skeleton via the **Pelvic** and **Shoulder Girdles**. The **Femur** is indeed the longest bone. The forearm has two bones: **Radius + Ulna**. Adults have **206** bones (not 300).

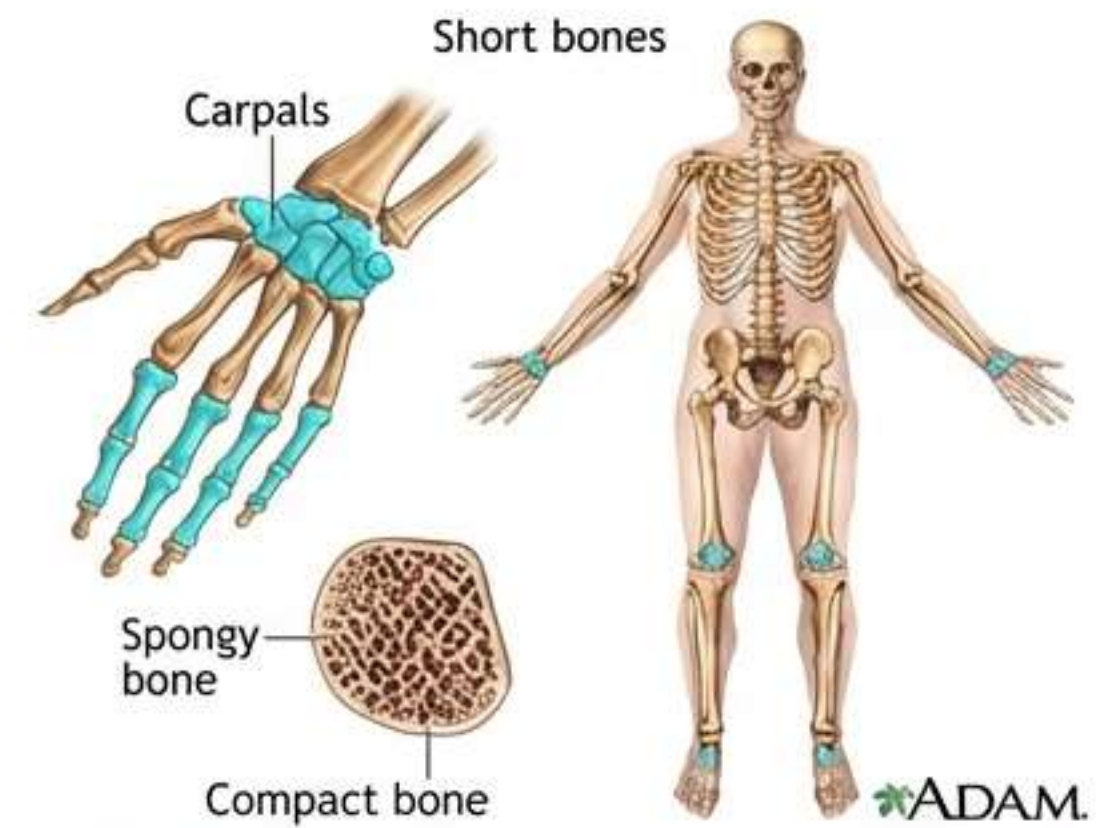
Girdles are the **Glue** (Highlighted in **Growth Green**). Forearm = **R.U.** ready? (Radius + Ulna).

QUESTION 12 | SOURCE: EMD 1 2020 (Q06)

About osteology:

- A. The skeleton is divided into axial and appendicular
- B. The scapula is a long bone
- C. The carpal bones are short bones
- D. The phalanges are long bones
- E. Bone is an elastic organ compared to cartilage

Correct Answer: A, C, D



Bones are classified by shape, not just size. **Phalanges** (fingers) are 'long bones' because they have a shaft and two heads, despite being small. **Carpals** (wrist) are 'short bones' (cuboid). The **Scapula** is a flat bone.

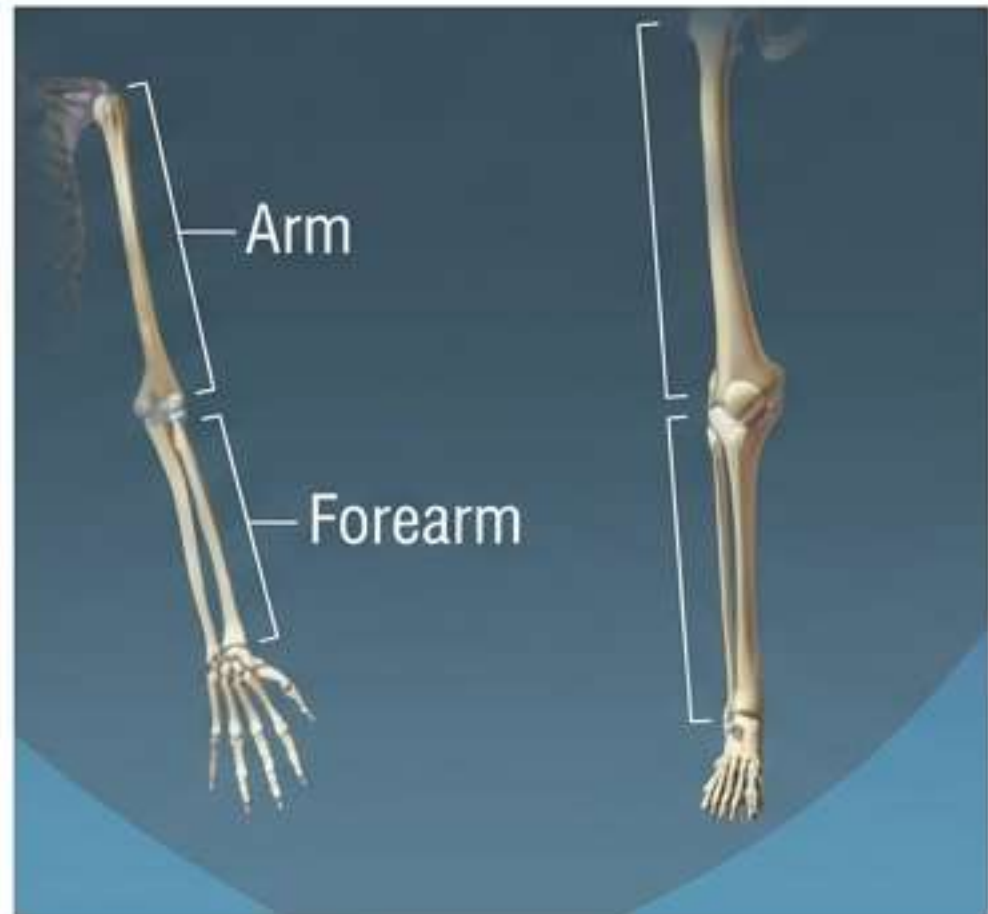
Phalanges are **Mini-Longs** (Highlighted in purple). **Scapula** is **Flat** (like a shovel).
By dentiste web

QUESTION 13 | SOURCE: EMD 1 2019 (Q01)

These propositions concern the skeleton:

- A. There are 300 constant bones in the human body
- B. Bone is richly vascularized and non-innervated
- C. The ulna alone constitutes the skeleton of the arm
- D. Long bones comprise a diaphysis and an epiphysis
- E. The human skeleton is divided into axial and appendicular skeleton

Correct Answer: E



Adults have 206 bones (A is false). Bone is innervated (B is false). The Arm contains the **Humerus**; the Ulna is in the *forearm* (C is false). While D is anatomically true, the exam key prioritizes **E** as the comprehensive classification fact.

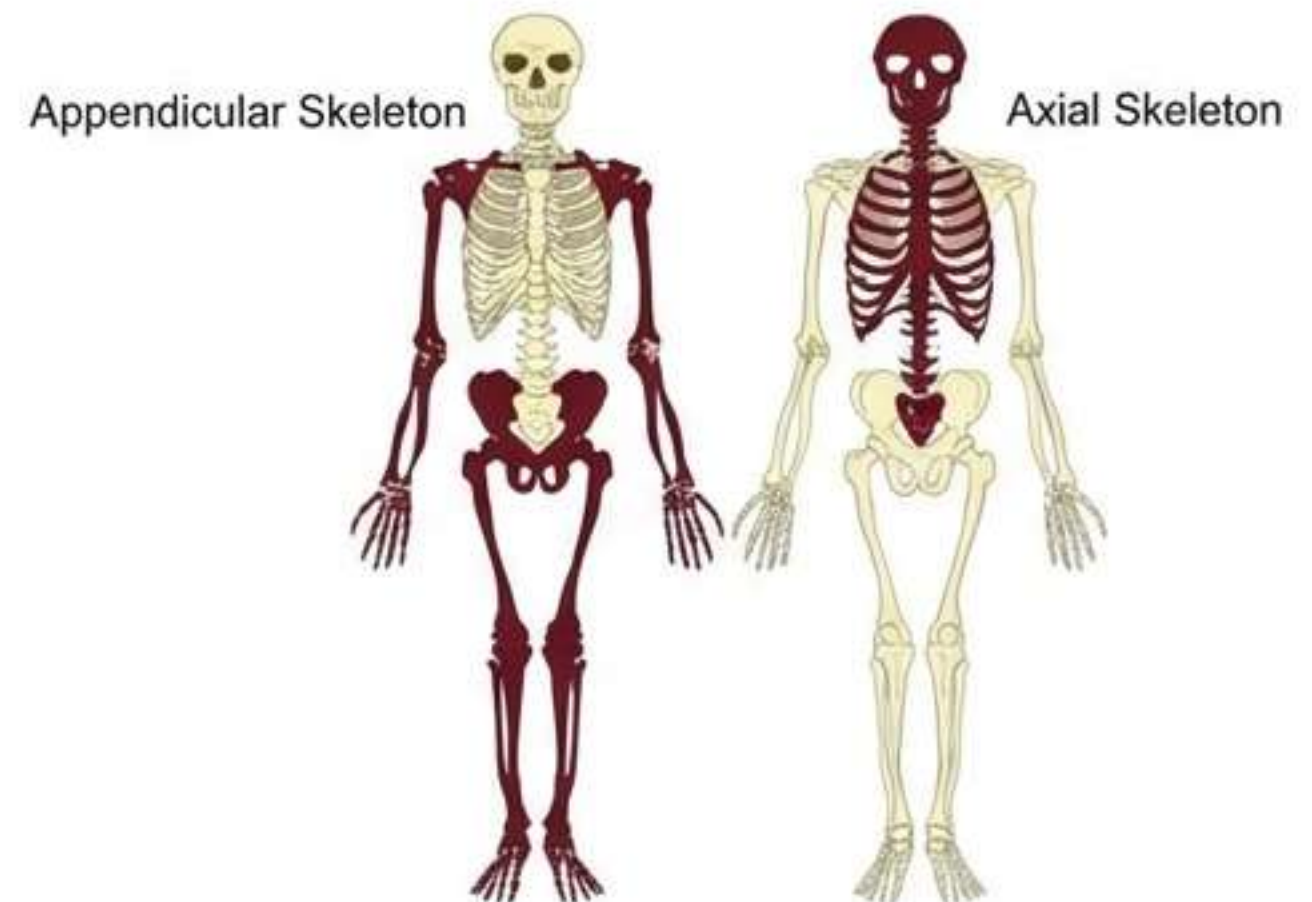
Arm vs Forearm. Arm = Humerus (Top). Forearm = Radius/Ulna (Bottom).

QUESTION 14 | SOURCE: EMD 1 2018 (Q01)

These propositions concern the skeleton:

- A. The human skeleton is divided into axial and appendicular skeleton
- B. The scapula forms the skeleton of the shoulder
- C. Bones comprise articular surfaces not covered by cartilage
- D. Long bones comprise a diaphysis and an epiphysis
- E. The ulna forms the skeleton of the arm

Correct Answer: A



Similar to Q13, **A** is the undeniable fundamental fact. **C** is false (articular surfaces *must* have cartilage). **E** is false (Ulna = forearm). The Scapula is *part* of the shoulder girdle, but the definition in **A** is the most robust answer.

Axial = **A**xis. **A**ppendicular = **A**ppendages (Limbs).

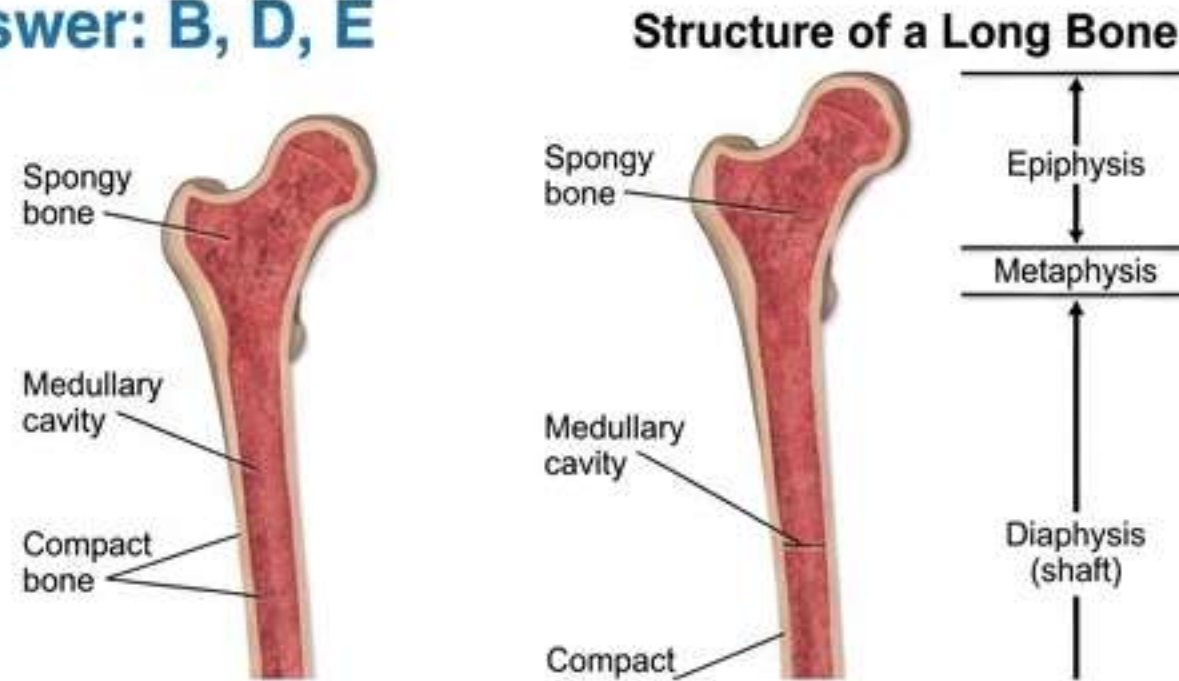
By dentiste web

QUESTION 15 | SOURCE: EMD 1 2017 (Q01)

A long bone presents the following characteristics:

- A. Equal length and width
- B. Length predominant over width
- C. Presents two faces and two edges
- D. A diaphysis and two extremities or epiphyses
- E. The humerus bone is the typical example

Correct Answer: B, D, E



The classic definition of a **Long Bone**:

1. Dimension: **Length** > Width.
2. Parts: **Diaphysis** (shaft) + **Epiphyses** (ends).
3. Example: **Humerus**, Femur, Tibia.

L.D.E.

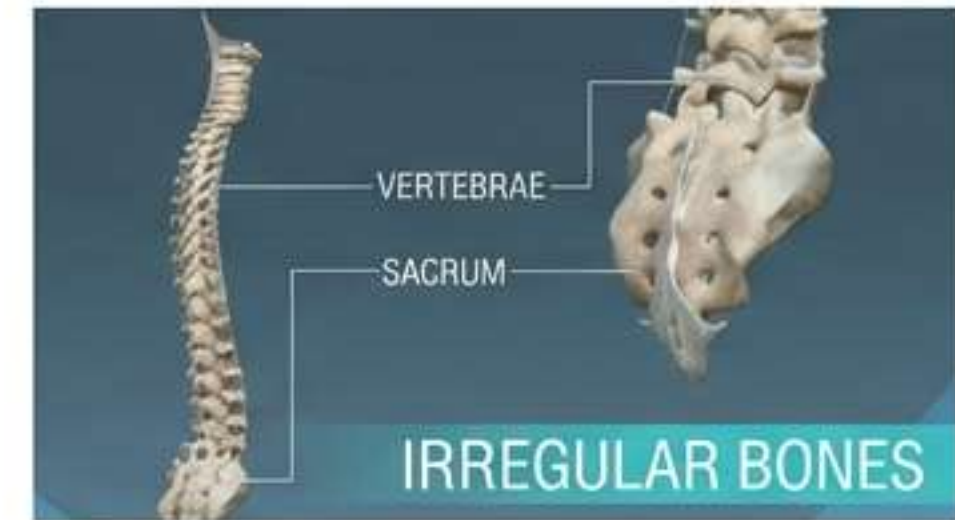
Length, **D**iaphysis, **E**piphysis.

QUESTION 16 | SOURCE: EMD 1 2017 (Q08)

These propositions concern the human skeleton:

- A. Consists of 600 bones
- B. The girdles connect the axial skeleton to the appendicular skeleton
- C. The flat bone has a length superior to the width
- D. Diaphysis and epiphysis are terms specific to long bone
- E. The thorax is part of the axial skeleton

Correct Answer: B, D, E



There are ~600 **muscles**, but only 206 **bones** (A is false). **Girdles** are the attachment points. Terms like 'Diaphysis' apply *only* to long bones (you don't say a vertebra has a diaphysis). Thorax (ribs/sternum) is central, thus **Axial**.

206 Bones.

600 Muscles.

Don't mix the numbers!

By dentiste web

QUESTION 17 | SOURCE: EMD 1 2016 (Q09)

A long bone is a bone that has the following characteristics:

- A. Equal length and width
- B. Length predominant over width
- C. Presents two faces and two extremities
- D. A body and two extremities or epiphyses
- E. The humerus bone is the typical example

Correct Answer: B, D, E



This question reinforces the core concept seen in Q15. Repetition is key to mastery. Note that **"Body"** is synonymous with **"Diaphysis"** in this context. The **Humerus** remains the standard textbook example.

Body = Shaft = Diaphysis.
Three words, one part.

End of Section



كلية طب الأسنان

Faculty Of Dental Medicine

- ✓ Bone Types
- ✓ Axial vs Appendicular
- ✓ Bone Structure
- ✓ Exam Traps